EC-341 Digital Systems Design
Experiment # 5

Implementation of Combinational Circuits - Adders

Objectives:

- Combinational Circuits
- The Adders
- Lab Tasks

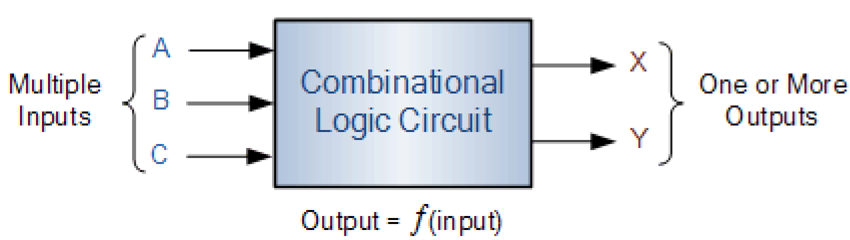
1. Combinational Circuits

Combinational Circuits (CC) are circuits made up of different types of logic gates. A logic gate is a basic building block of any electronic circuit. The output of the combinational circuit depends on the values at the input at any given time. The circuits do not make use of any memory or storage device. Some of the most common types of combinational circuit includes adder, subtractor, multiplexer and demultiplexer.

Unlike Sequential Logic Circuits whose outputs are dependent on both their present inputs and their previous output state giving them some form of Memory. The outputs of Combinational Logic Circuits are only determined by the logical function of their current input state, logic “0” or logic “1”, at any given instant in time.

The result is that combinational logic circuits have no feedback, and any changes to the signals being applied to their inputs will immediately have an effect at the output. In other words, in a Combinational Logic Circuit, the output is dependant at all times on the combination of its inputs. Thus a combinational circuit is memoryless.

So if one of its inputs condition changes state, from 0-1 or 1-0, so too will the resulting output as by default combinational logic circuits have “no memory”, “timing” or “feedback loops” within their design.

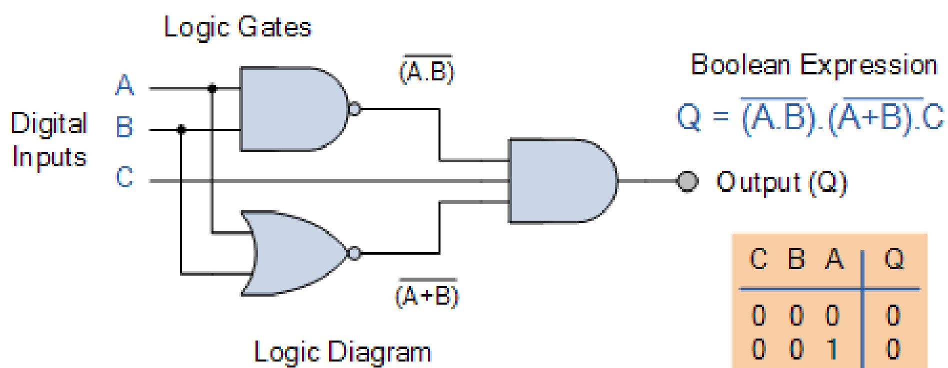


Combinational Logic Circuits are made up from basic logic NAND, NOR or NOT gates that are "combined" or connected together to produce more complicated switching circuits. These logic gates are the building blocks of combinational logic circuits. An example of a combinational circuit is a decoder, which converts the binary code data present at its input into a number of different output lines, one at a time producing an equivalent decimal code at its output.

The three main ways of specifying the function of a combinational logic circuit are:

- Boolean Algebra – This forms the algebraic expression showing the operation of the logic circuit for each input variable either True or False that results in a logic "1" output.
- Truth Table – A truth table defines the function of a logic gate by providing a concise list that shows all the output states in tabular form for each possible combination of input variable that the gate could encounter.
- Logic Diagram – This is a graphical representation of a logic circuit that shows the wiring and connections of each individual logic gate, represented by a specific graphical symbol, that implements the logic circuit.

All three of these logic circuit representations are shown below.



Typical Truth Table

C	B	A	Q
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

As combinational logic circuits are made up from individual logic gates only, they can also be considered as "decision making circuits" and combinational logic is about combining logic gates together to process two or more signals in order to produce at least one output signal according to the logical function of each logic gate. Common combinational circuits made up from individual logic gates that carry out a desired application include Multiplexers, De-multiplexers, Encoders, Decoders, Full and Half Adders etc.

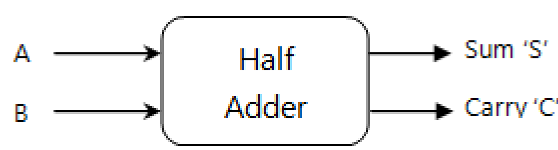
2. The Adders

An adder is a digital circuit that is used to perform the addition of numeric values. It is one of the most basic circuits and is found in arithmetic logic units of computing devices. There are two types of adders. Half adders compute single digit numbers, while full adders compute larger numbers.

2.1 Half Adder

The half adder adds two single digit binary numbers and forms the foundation for all addition operations in computing. If we have two single binary digits, A and B, then the half adder adds them with the circuit carrying two outputs, the sum and the carry. The carry represents any overflow from the addition of the two numbers. This is represented in the following block diagram figure:

Figure 1: Half Adder



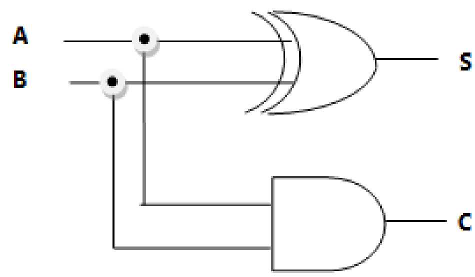
In addition, the following truth table demonstrates all the possible outputs for various input combinations of the half adder.

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Table 1: Truth Table - Half Adder

The next figure represents the logic circuit of the half adder:

Figure 2: Half Adder- Logic Circuit

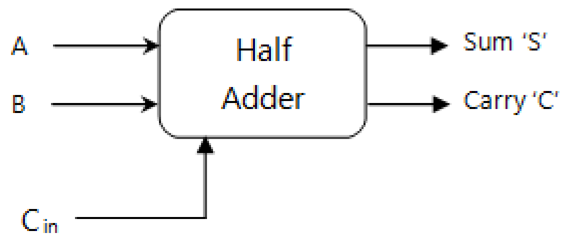


The sum S is represented by the Boolean Expression S = A'B + AB' and C = AB

2.2 Full Adder

The full adder overcomes the disadvantages of the half adder in that it can add two single bit numbers in addition to the carry digit at its input as seen in this figure:

Figure 3: Full Adder



The next truth table shown here demonstrates all the possible outputs for various input combinations with the carry input digit:

A	B	Cin	Co	S
0	0	0	0	0
0	0	1	0	1

0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	1
1	1	1	1	1

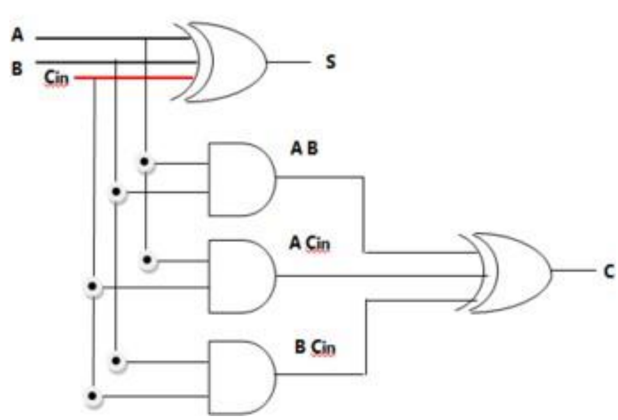
Boolean expression for the full adder is S = A'B'Cin + A'BCin' + AB'Cin' + ABCin and C = A'BCin + AB'Cin + ABCin' + ABCin. This is where A and B are all the possible binary inputs and C is the carry in. For example, if A is 0 and B is 0 and the Cin is 1, then:

S = (0'0'1)+(0'01')+(00'1')+(001) = (111)+(100)+(010)+(001) = (1)+(0)+(0)+(0) = 1

C = (0'01)+(00'1)+(001')+(001) = (101)+(011)+(000)+(001) = (0)+(0)+(0)+(0) = 0

S = 1 and C = 0

Figure 3: Full Adder- Logic Circuit



3. Tasks

1. Write a Behavioral Level Verilog Code for Half Adder. Run the code on ZYBO and verify results using different instances.
2. Write a Behavioral Level Verilog Code for Full Adder. Run the code on ZYBO and verify results using different instances.
3. Write a Behavioral Code for 2bit Adder. Run the code on ZYBO and verify results using different instances.
4. Write a Behavioral Level Verilog Code for 2 bit 2's Complement. Run the code on ZYBO and verify results using different instances.